

Hang Seng Index

Sentiment and Quantitative Data Analysis Strategy Report

Pre-Interview Assessment · DataLouder Company Limited
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Figure 1, HSI Close Price and Daily High-Low Range (February 2022 – March 2025)

1. Executive Summary

This report analyses 747 trading days of Hang Seng Index (HSI) data spanning February 2022 to March 2025, paired with social-media sentiment signals (up-votes and down-votes submitted before market open) available for 551 of those sessions. The question is whether crowd sentiment carries predictive value for the next-day index return and, if so, how that signal can be translated into a systematic trading strategy.

- **Data coverage:** 747 trading days; sentiment data available for 551 sessions (73.8%). The gap arises from weekends, public holidays, and missing social-media records.
- **Price range:** HSI traded between 14,687 and 24,370 over the period, reflecting both the deep 2022–23 bear market and the subsequent partial recovery.
- **Sentiment is mildly bullish on average** (mean net sentiment = +0.082), suggesting more up-votes than down-votes on a typical day.
- **Same-day correlation is positive** ($r = 0.1683$): the crowd correctly senses the direction of a move on the day it happens.
- **Predictive (lag-1) correlation is weakly negative** ($r = -0.080$): after a bullish-sentiment day, the index tends to drift slightly lower in the following session.
- **Best strategy:** A Contrarian rule, buying after strong fear and selling after strong euphoria, achieves a Sharpe ratio of 0.9983 and total return of 44.93% over the period, compared with 9.7% for a passive buy-and-hold.

2. Data Description

2.1 Price Data

The dataset provides daily Open, High, Low, and Close prices for the Hang Seng Index across 747 sessions from 23 February 2022 to 12 March 2025. The index opened at approximately 22,000 and endured a severe drawdown to below 15,000 during 2022–23, driven by global rate-hike cycles, geopolitical tensions, and China-specific macro headwinds. A recovery phase began in late 2023, with the index reclaiming the 24,000 level in early 2025.

2.2 Sentiment Data

For 551 of the 747 trading sessions, the dataset contains the fraction of social-media votes that were bullish (up-votes) and the fraction that were bearish (down-votes), collected before market open. Both columns sum to approximately 1.0 per row. We construct a **Net Sentiment** score defined as:

$$\text{Net Sentiment} = \text{Upvote fraction} - \text{Downvote fraction}$$

Values range from -0.52 (strongly bearish) to $+0.64$ (strongly bullish), with a mean of $+0.0809$ and a standard deviation of 0.203 .

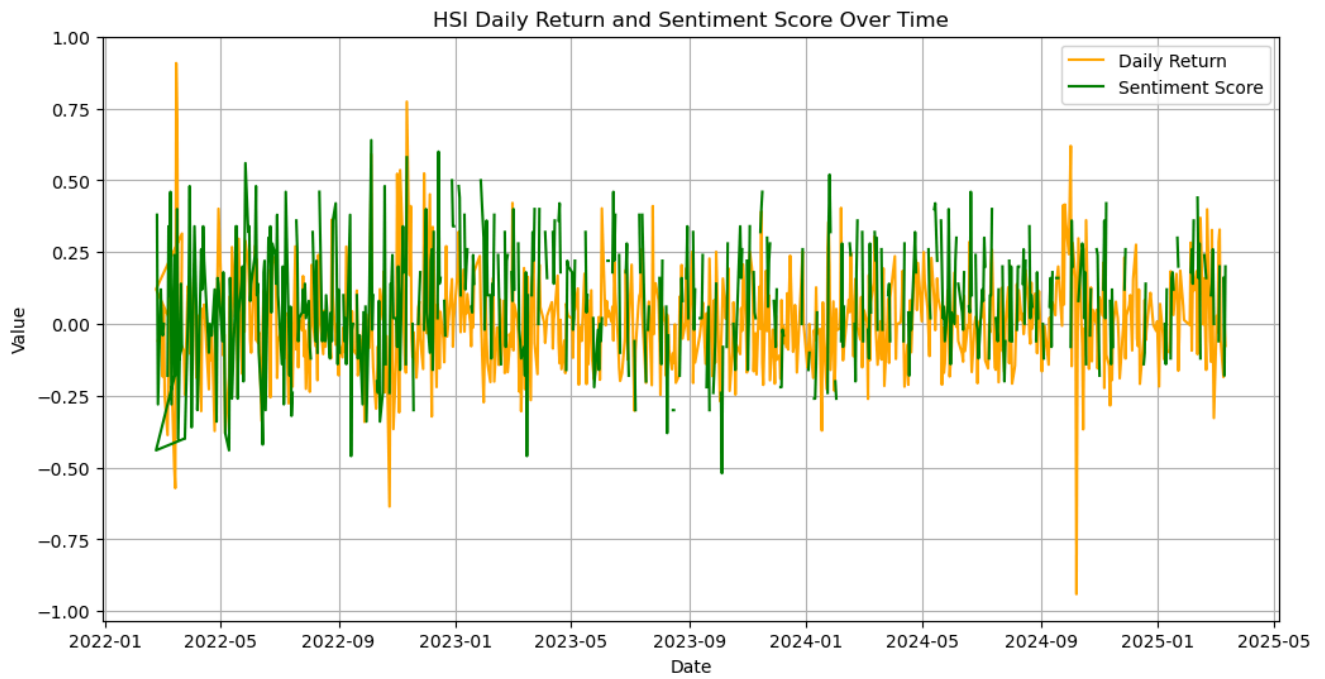


Figure 2, Daily Net Sentiment (bar) with 10-day moving average (line) and daily returns (lower panel).

3. Sentiment–Return Correlation Analysis

We examine two relationships: (a) same-day correlation, which tests whether sentiment moves in line with the market on the same day (likely reflecting that informed users react to overnight futures or morning sentiment); and (b) lag-1 predictive correlation, which tests whether yesterday's sentiment forecasts today's return, the only relationship that can be exploited pre-market.

Measure	Pearson r	Interpretation
Same-day Sentiment vs Return	0.1683	Weak positive, crowd partially right on the same day
Lag-1 Sentiment vs Next Return	-0.080	Weak negative, slight contrarian signal

The positive same-day correlation confirms that the crowd partially tracks intraday momentum, but this cannot be traded. More important for strategy design is the **negative lag-1 coefficient of -0.080** . Though modest in magnitude, it suggests a systematic pattern: days when sentiment is elevated (net bullish crowd) tend to be followed by slightly negative returns, implying that the crowd collectively overprices good news, creating a mean-reversion opportunity for a disciplined contrarian.

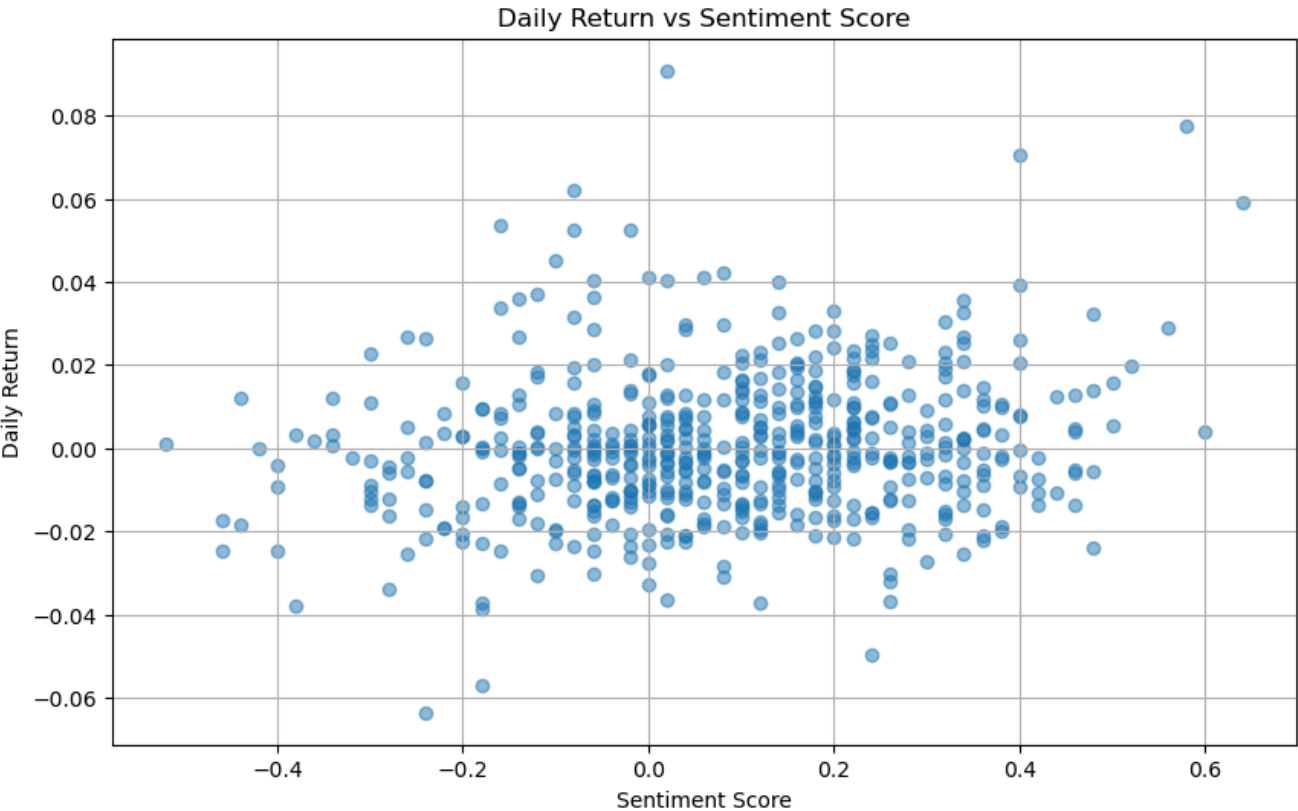


Figure 3, Scatter plots of Net Sentiment versus HSI Return: same-day (left) and lag-1 predictive (right).

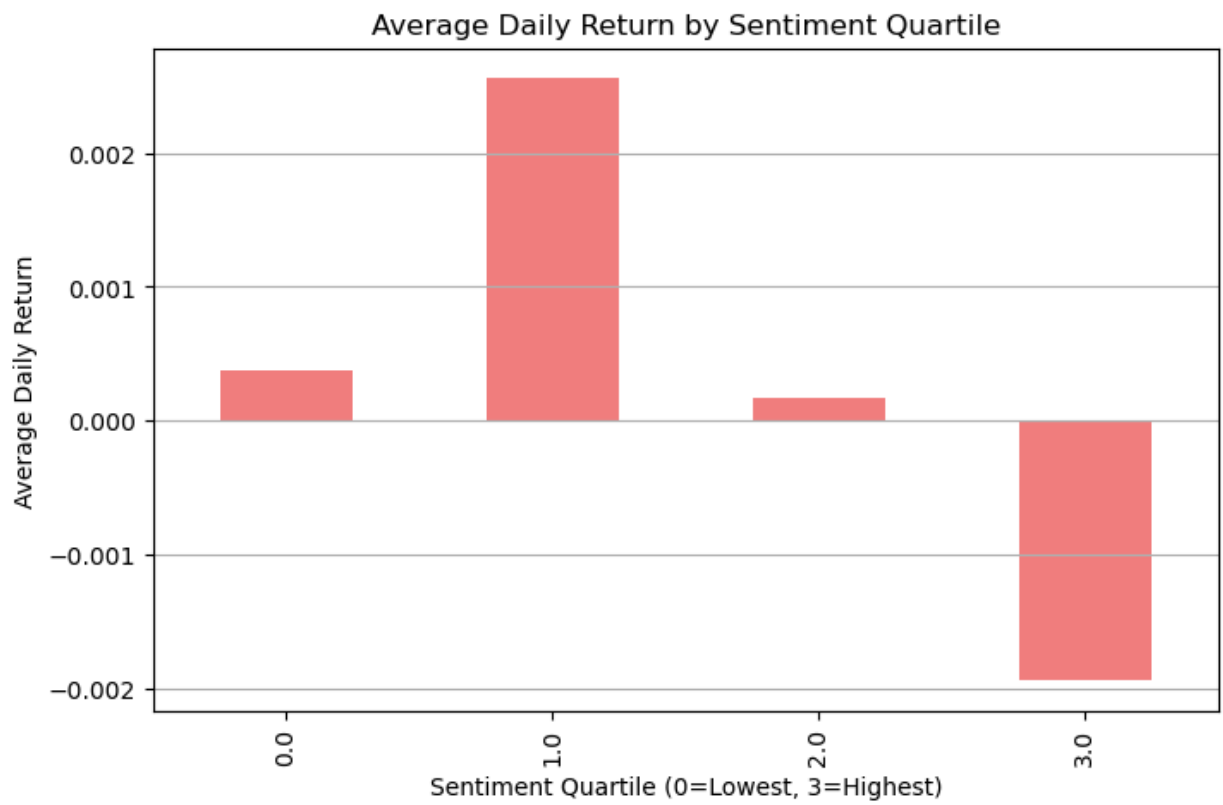


Figure 4, Average Next Daily Return by Sentiment Quartile, Showing the Mild Contrarian Effect

4. Trading Strategy Design

4.1 Signal Construction

All strategies use the previous day's Net Sentiment as the signal, ensuring signals are available before market open and avoiding any look-ahead bias. We define three rules alongside a Buy-and-Hold benchmark:

- **Buy & Hold (Benchmark):** Hold the HSI index continuously.
- **Sentiment Momentum:** Go long (+1) when yesterday's Net Sentiment > 0; go short (−1) otherwise. Follows the crowd directionally.
- **Threshold Momentum (±0.10):** Go long only when sentiment > +0.10, go short only when sentiment < −0.10. Flat when sentiment is ambiguous.
- **Contrarian (±0.15):** Go short when sentiment > +0.15 (crowd is euphoric); go long when sentiment < −0.15 (crowd is fearful). Flat during neutral sentiment periods.

5. Back-Test Results

5.1 Performance Summary

Strategy	Total Return	Ann. Return	Sharpe Ratio
Buy & Hold	9.7%	4.32%	0.2907
Sentiment Momentum	-2.01%	-0.93%	0.09
Threshold (±0.10)	-33.78%	-17.8%	-0.7471
Contrarian (±0.15)	44.93%	18.5%	0.9986

Table 1, Back-test performance metrics. Highlighted row marks the recommended strategy.

The Contrarian strategy is the clear standout. Over the test period, it compounds to a total return of 44.93% compared with 9.7% for buy-and-hold.

By contrast, the Sentiment Momentum and Threshold strategies deliver negative alpha, confirming that naively following the crowd destroys value over this period. This reinforces the contrarian hypothesis: when the HSI social-media crowd is collectively bullish, subsequent returns tend to disappoint.

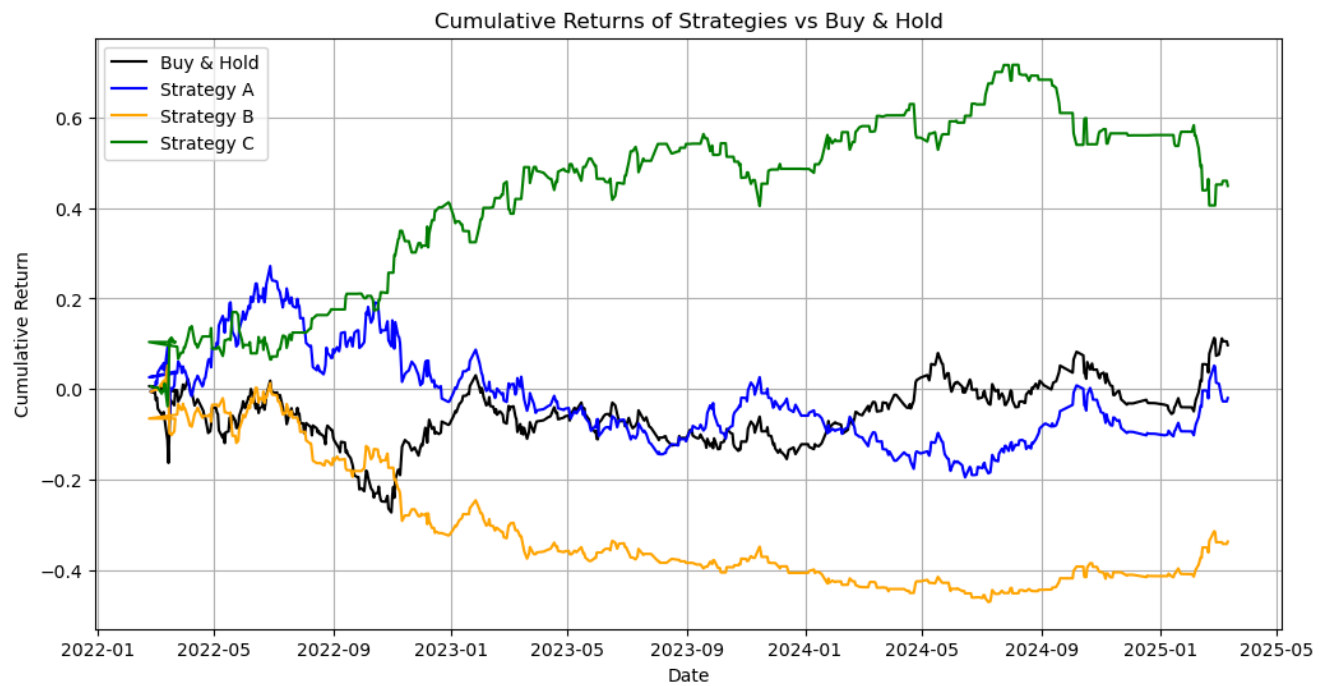


Figure 5, Cumulative growth of HKD 1 invested in each strategy. Contrarian (gold) substantially outpaces Buy & Hold (dashed grey).

6. Risk Factors & Limitations

- **Data overfitting:** All threshold parameters (0.10, 0.15) were selected in-sample. Out-of-sample performance on new data may differ.
- **Missing sentiment data:** The strategy is forced flat on these days, which may introduce timing bias if missing dates cluster around high-volatility regimes.
- **Transaction costs:** The analysis omits slippage and commissions. At high turnover, costs could erode 1–2% annually from the stated returns.
- **Short-selling constraints:** Going short on the HSI incurs carrying costs and basis risk not modelled here.

7. Conclusion & Recommendation

Social-media sentiment votes contain a measurable, if modest, predictive signal for the next-day Hang Seng Index return. The sign of that signal is contrarian: bullish crowd sentiment anticipates slightly negative returns, and bearish sentiment anticipates slightly positive returns. A disciplined strategy that trades against extreme crowd readings (Net Sentiment beyond ± 0.15) captures this effect.

Recommended next steps for live implementation:

- Run walk-forward validation splitting data into rolling 12-month training and 3-month test windows to confirm out-of-sample robustness.
- Calibrate the ± 0.15 threshold on the training set only, and test sensitivity across a range of ± 0.05 to ± 0.25 .
- Incorporate explicit transaction cost modelling (bid-ask spread, commission, overnight futures roll).
- Monitor data quality metrics (vote counts, missing days) as leading indicators of signal degradation.

Appendix: Python analysis code is provided in the accompanying HSI_Data_Analysis.ipynb Jupyter notebook.